Sparse Tensor Attention for Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing (HU) is a fundamental task for resolving the mixed pixel problem by decomposing a hyperspectral image (HSI) into constituent endmembers and their corresponding abundances. Although deep-learning-based HU methods have achieved promising performance, most rely on manually designed network architectures that encode spatial–spectral priors only implicitly, limiting their interpretability and ability to model complex spatial–spectral characteristics. To address this limitation, we propose a self-supervised sparse tensor attention (TA) framework for HU, termed STU. The proposed method builds upon the mathematical connection between self-attention and Tucker decomposition and exploits the nonlocal spatial–spectral self-similarity and low-rank structure inherent in HSIs. Specifically, pairwise dependencies among local spatial–spectral cubes are encoded into a sparse tensorized attention core, which is projected through mode-wise low-rank factor matrices to recover the abundance representations. The resulting abundances are constrained to satisfy the abundance nonnegativity and sum-to-one constraints and are subsequently combined with a learnable endmember matrix under the linear mixture model (LMM) to reconstruct the observed HSI. This formulation integrates data-adaptive nonlocal dependency modeling, multilinear low-rank representation, and physically consistent endmember-abundance estimation within a unified framework. Extensive experiments on synthetic and real-world hyperspectral datasets demonstrate that STU achieves competitive or superior performance compared with representative model-based and deep-learning methods.**

***Index Terms*—Hyperspectral unmixing (HU), nonlocal self-similarity, self-attention, tensor decomposition.**

I. Introduction

HYPERSPECTRAL imaging simultaneously captures 2-D spatial information and 1-D spectral information across hundreds of contiguous narrow bands [1], making it a powerful remote sensing technique. Benefiting from its rich spectral information, hyperspectral imagery has demonstrated considerable effectiveness in diverse applications, including object tracking [2], [3] and pixel classification [4], [5]. However, because of the limited spatial resolution of hyperspectral sensors and the effects of atmospheric scattering, the spectrum observed at a single pixel is often a mixture of several constituent materials [6], [7]. Hyperspectral unmixing (HU) addresses this mixed-pixel problem by decomposing each observed spectrum into a set of pure spectral signatures, referred to as endmembers, and their corresponding fractional abundances [8].

The physical spectral mixing process is generally described using either linear or nonlinear mixture models [9]. Nonlinear mixture models account for multiple reflections, scattering, or interactions among different materials before the incident radiation reaches the sensor. In contrast, the linear mixture model (LMM) assumes that the observed spectrum is formed by a linear combination of the constituent endmember spectra. Owing to its physical interpretability and mathematical simplicity, the LMM remains the most widely adopted formulation for HU.

LMM-based unmixing methods can generally be divided into supervised, semi-supervised, and unsupervised approaches [10]. Supervised methods focus primarily on abundance estimation under the assumption that the endmembers are known a priori. In practice, HU is often conducted sequentially, with endmember extraction followed by abundance estimation. Geometric methods are particularly prominent for endmember extraction. These methods represent hyperspectral pixels as points in a high-dimensional simplex and identify the simplex vertices as endmembers. Representative algorithms include vertex component analysis (VCA) [11] and N-FINDR [12]. To address scenes in which pure pixels are unavailable, minimum-volume-based methods [13], [14], [15], [16] seek the minimum-volume simplex that encloses the observed data. Given the estimated endmembers, the fully constrained least-squares algorithm (FCLS) [17] and its computationally efficient variants [18] are widely used to estimate abundances subject to the abundance nonnegativity and sum-to-one constraints. Semi-supervised methods select an appropriate subset of endmembers from a predefined spectral library. Under this setting, unmixing is commonly formulated as a sparse-regression problem that balances reconstruction fidelity against the sparsity of the selected endmembers [19], [20].

In unsupervised HU, both the endmembers and their corresponding abundances are unknown and must be estimated jointly. Nonnegative matrix factorization (NMF) [21] has become a representative approach because its parts-based

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representation naturally accords with the nonnegativity of endmembers and abundances. Nevertheless, the standard NMF objective is nonconvex and may converge to undesirable local optima. To improve solution stability, various regularization terms, including abundance sparsity [22], [23] and spatial smoothness [24], [25], [26], have been incorporated into NMF-based models. However, NMF requires the 3-D hyperspectral image (HSI) cube to be unfolded into a 2-D matrix, thereby treating pixels as independent spectral vectors and weakening the modeling of spatial neighborhood information. Tensors recently show more capacity for multidimensional data modeling [27], [28], [29]. Nonnegative tensor factorization (NTF) [30] extends NMF to higher-order representations and operates directly on the HSI in its native tensor form, thereby better preserving intrinsic spatial–spectral correlations. In particular, Qian et al. [30] established a connection between NTF and the LMM and proposed matrix–vector NTF (MV-NTF). MV-NTF decomposes an HSI into a sum of component tensors, each represented as the outer product of an endmember vector and its corresponding abundance matrix. By exploiting low-rank structures in both the spatial and spectral domains, MV-NTF achieves effective and physically interpretable unmixing.

Traditional model-based approaches, however, are often limited by restrictive assumptions, sensitivity to observation noise [31], and insufficient representation capacity. Consequently, deep-learning-based methods have received increasing attention in HU. Among them, autoencoders have emerged as a dominant framework because their latent representations can be interpreted as abundance maps, while the decoder weights can be associated with endmember signatures [32], [33], [34], [35], [36], [37]. In this framework, the encoder serves as an inference network that predicts abundance features, and the entire network is optimized by minimizing the reconstruction error between the observed and reconstructed HSIs. The architecture of the encoder is therefore critical because it implicitly determines the data-driven prior used to regularize abundance estimation. For example, Gao et al. [38] employed cascaded autoencoders to reduce the loss of material information during reconstruction. To overcome the limited receptive fields of convolutional neural networks, Transformer-based approaches have been introduced to model long-range dependencies in HSIs. Wang et al. [39] developed a dual-aware encoder to capture both long-range spectral correlations and global spatial context. Yang et al. [40] incorporated shifted-window mechanisms into a U-shaped architecture to recover spatially accurate abundance maps with rich multiscale nonlocal information. Su et al. [41] proposed SRViT, which uses relation-aware self-attention and relative positional biases to model spatial relationships explicitly. Deep-unfolding methods have also been developed to integrate physical unmixing models with the representation capability of neural networks by transforming iterative optimization procedures into learnable network architectures [42], [43].

Despite their promising performance, many existing deep unmixing methods rely on predetermined and heuristic architectures whose underlying priors are encoded only implicitly. Compared with classical sparse or low-rank priors, these architectural priors are often less interpretable and may not generalize consistently across different hyperspectral scenes. This limitation raises an important question: *Can an adaptive architecture be constructed directly from the intrinsic low-dimensional structure of HSIs to improve HU?*

Recently, Chen et al. [44] introduced a tensor-attention model for physical-field reconstruction by revealing a connection between Tucker decomposition and self-attention. In this interpretation, the Tucker core tensor acts as a tensorized weighting map, analogous to an attention map, while the factor matrices provide mode-wise value representations. The reconstructed tensor can therefore be viewed as a weighted combination of multilinear interactions among these factor matrices. Based on this insight, they developed a complexity-adaptive neural representation grounded in Tucker decomposition and demonstrated its universal approximation capability for continuous physical fields.

Motivated by these observations, we propose a self-supervised sparse tensor attention (TA) framework for HU, termed STU, as illustrated in Fig. 1. Specifically, the input HSI is first partitioned into a set of overlapping spatial–spectral cubes, which are treated as tokens. Pairwise similarities among these tokens are then computed to characterize nonlocal spatial–spectral dependencies. By applying sparse normalization, weak and irrelevant correlations are suppressed, yielding a sparse attention representation. Based on the mathematical correspondence between self-attention and Tucker decomposition, the attention coefficients are reorganized into tensorized cores that preserve the intrinsic 3-D arrangement of the HSI tokens. These cores are subsequently projected through low-rank factor matrices to recover structured abundance representations. This design jointly exploits local contextual information, nonlocal self-similarity, and multilinear low-rank structure, thereby providing effective regularization for abundance estimation. The estimated abundances are further coupled with a learnable endmember matrix under the LMM, enabling the joint recovery of physically meaningful abundances and endmembers. Extensive experiments on both synthetic and real-world hyperspectral datasets demonstrate that STU achieves competitive or superior performance in abundance estimation, endmember recovery, and robustness to noise compared with representative state-of-the-art methods.

The contributions of this article are summarized as follows.

1) We propose a novel self-supervised HU framework which formulates tensorized self-attention as a Tucker decomposition for abundance estimation. This formulation enables the local and nonlocal spatial–spectral dependencies of an HSI to be modeled within a unified low-rank tensor representation.
2) By further coupling the resulting abundance representation with a learnable endmember matrix under the LMM, the proposed framework jointly estimates physically meaningful abundances and endmembers while satisfying the abundance nonnegativity and sum-to-one constraints.
3) Extensive experiments on synthetic and real-world hyperspectral datasets demonstrate that STU consistently achieves competitive or superior performance compared with state-of-the-art unmixing methods. The results

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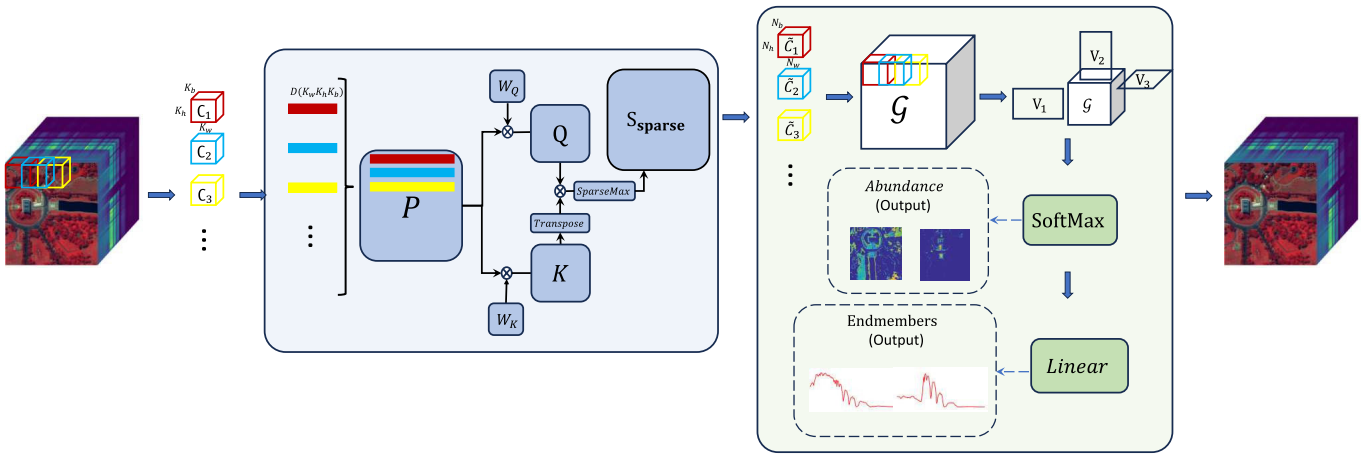


Fig. 1. Framework of the proposed STU method. Based on the connection between Tucker decomposition and self-attention, we establish a novel self-supervised network for HU.

confirm its effectiveness in abundance estimation, endmember recovery, and robustness under different mixing levels, image sizes, and noise conditions.

The remainder of this article is organized as follows. Section II introduces the background. Section III details the proposed method. Section IV presents the experimental results. Finally, Section V concludes this article.

II. Background

This section describes the LMM, Vision Transformer (ViT), and Tucker decomposition.

A. Linear Mixture Model

Consider an observed HSI represented as a third-order tensor $\mathcal{Y} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B indicate the spatial height, spatial width, and the number of spectral bands, respectively. Under the matrix formulation, the tensor is typically unfolded into a 2-D matrix $\mathbf{Y} \in \mathbb{R}^{B\times N}$, where $N = H \times W$ is the total number of pixels. The LMM assumes that the spectrum of each pixel is a linear combination of endmembers weighted by their corresponding abundances, modeled as

$$\mathbf{Y} = \mathbf{MA} + \mathbf{E}. \tag{1}$$

Here, $\mathbf{M} \in \mathbb{R}^{B\times P}$ is the endmember matrix containing P distinct spectral signatures, and $\mathbf{A} \in \mathbb{R}^{P\times N}$ is the abundance matrix representing the fractional contribution of each endmember to the pixels. $\mathbf{E} \in \mathbb{R}^{B\times N}$ represents the additive noise and measurement errors. To guarantee the physical validity of the unmixing process, the abundance fractions are subject to the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), defined as follows:

$$\mathbf{A} \geq 0, \quad \mathbf{1}_P^T\mathbf{A} = \mathbf{1}_N^T \tag{2}$$

where $\mathbf{A} \geq 0$ enforces nonnegativity for all entries, and $\mathbf{1}_P^T\mathbf{A} = \mathbf{1}_N^T$ ensures that the abundances for each pixel sum to one.

B. Vision Transformer

ViT [45] treats an image as a sequence of patches to capture long-range dependencies. The core component, self-attention, operates on input features $\mathbf{X} \in \mathbb{R}^{N\times C}$, where N is the number of tokens and C is the channel dimension. The input is linearly projected into query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices

$$\mathbf{Q} = \mathbf{XW}^Q, \quad \mathbf{K} = \mathbf{XW}^K, \quad \mathbf{V} = \mathbf{XW}^V \tag{3}$$

where $\mathbf{W}^Q, \mathbf{W}^K$, and $\mathbf{W}^V$ are learnable weight matrices. The attention weights are computed using the scaled dot-product

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{SoftMax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right)\mathbf{V}. \tag{4}$$

Here, $\sqrt{d_k}$ is a scaling factor. The Softmax function performs row normalization on the scaled dot-product $\mathbf{QK}^T$. This operation results in an attention map of size $N \times N$, where the nth row represents the similarities between the nth token and all other tokens. The self-attention mechanism functions as an expansion of the input features in terms of the value vectors, with weighting coefficients expressing their mutual similarities.

C. Tucker Decomposition Model

Let $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ be an Nth-order tensor. The mode-n product of a tensor $\mathcal{X}$ with a matrix $\mathbf{V} \in \mathbb{R}^{J\times I_n}$, denoted by $\mathcal{X} \times_n \mathbf{V}$, results in a tensor of size $I_1 \times \cdots \times J \times \cdots \times I_N$. Element-wise, it is expressed as

$$(\mathcal{X} \times_n \mathbf{V})_{i_1\ldots j\ldots i_N} = \sum_{i_n=1}^{I_n} x_{i_1\ldots i_n\ldots i_N} \cdot v_{ji_n}. \tag{5}$$

Based on this operation, the Tucker decomposition factorizes a tensor into a core tensor multiplied by a factor matrix along each mode. Specifically, for a third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$, the Tucker model decomposes it into a core tensor $\mathcal{G}$ and three factor matrices $\mathbf{V}_1, \mathbf{V}_2$, and $\mathbf{V}_3$

$$\mathcal{X} = \mathcal{G} \times_1 \mathbf{V}_1 \times_2 \mathbf{V}_2 \times_3 \mathbf{V}_3 \tag{6}$$

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where $\mathcal{G} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ is the core tensor capturing the interaction between different modes, and $\mathbf{V}_n \in \mathbb{R}^{I_n \times R_n}$ (for $n = 1, 2, 3$) are the factor matrices. The tuple (R_1, R_2, R_3) represents the multilinear rank of the tensor.

It should be noted that when setting $\mathbf{V}_1$ and $\mathbf{V}_2$ as identity matrices and $\mathbf{V}_3$ to $\mathbf{M}$, the Tucker decomposition becomes

$$\mathcal{Y} = \mathcal{G} \times_3 \mathbf{M}. \tag{7}$$

This is in essence the tensor version of the LMM where $\mathcal{G} \in \mathbb{R}^{H \times W \times P}$ is the abundance tensor.

III. Proposed Method

In this section, we present the details of our STU method. We first show the connection between Tucker decomposition and self-attention. Based on this connection, we construct a new framework for the HU task. The implementation details are also discussed.

A. Connection Between Tucker Decomposition and Attention

In (4) and (6), self-attention and Tucker decomposition are both weighted representation mechanisms. Their main difference lies in the structure of the basis elements being aggregated. Standard self-attention represents each output token as a weighted combination of vector features, whereas Tucker decomposition represents an output tensor as a weighted combination of multilinear basis elements. This shared weighted-expansion structure provides a natural mathematical connection between the two formulations. The connection between Tucker decomposition and self-attention is illustrated in Fig. 2.

For standard self-attention, the normalized attention matrix is defined as

$$\mathbf{S} = \text{SoftMax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}}\right) \in \mathbb{R}^{N \times N}. \tag{8}$$

The coefficient $S_{x,y}$ measures the dependency between the xth query token and the yth key token. Accordingly, the output associated with the xth query token is

$$\mathbf{o}_x = \sum_{y=1}^{N} S_{x,y} \mathbf{v}_y. \tag{9}$$

The above equation shows that self-attention is a coefficient-weighted expansion in which the attention coefficients $S_{x,y}$ determine the contributions of the value vectors $\mathbf{v}_y$.

A third-order Tucker model has an analogous multilinear expansion. The element-wise form of (6) is

$$\begin{aligned}\mathcal{X}(i_1, i_2, i_3) = &\sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \sum_{r_3=1}^{R_3} \mathcal{G}(r_1, r_2, r_3) \\ &\times V_1(i_1, r_1) V_2(i_2, r_2) V_3(i_3, r_3). \end{aligned} \tag{10}$$

Equivalently, (10) can be written as a weighted sum of rank-one tensor bases

$$\mathcal{X} = \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \sum_{r_3=1}^{R_3} \mathcal{G}(r_1, r_2, r_3) \times \left[\mathbf{v}_{1,r_1} \circ \mathbf{v}_{2,r_2} \circ \mathbf{v}_{3,r_3}\right] \tag{11}$$

where $\circ$ denotes the outer product, and $\mathbf{v}_{l,r_l}$ denotes the r_lth column of $\mathbf{V}_l$. Therefore, the entries of the Tucker core act as weighting coefficients, while the outer products of the factor vectors constitute tensor-valued basis elements. This formulation is structurally analogous to (9). The difference is that conventional self-attention aggregates vector-valued features, whereas Tucker decomposition aggregates separable tensor-valued features.

We next establish this correspondence in the context of an HSI. Multiple overlapping spatial–spectral cubes are extracted using a window size (K_h, K_w, K_b) and a stride (S_h, S_w, S_b). The numbers of cubes along the three dimensions are calculated as $N_h = \lfloor (H - K_h)/(S_h) \rfloor + 1$, $N_w = \lfloor (W - K_w)/(S_w) \rfloor + 1$, $N_b = \lfloor (B - K_b)/(S_b) \rfloor + 1$. The total number of tokens is therefore $N = N_h N_w N_b$. Each extracted cube $\mathcal{C}_n \in \mathbb{R}^{K_h \times K_w \times K_b}$ is flattened into a vector, producing the token matrix $\mathbf{P} \in \mathbb{R}^{N \times D}$, where $D = K_h K_w K_b$.

Because the tokens are sampled from a 3-D grid, each 1-D token index has a unique spatial–spectral coordinate. We define the bijective index mapping as

$$\pi(j_h, j_w, j_b) = (j_h - 1) N_w N_b + (j_w - 1) N_b + j_b \tag{12}$$

where $1 \le j_h \le N_h$, $1 \le j_w \le N_w$, and $1 \le j_b \le N_b$.

For a fixed query token x, the xth row of the attention matrix, $\mathbf{s}_x = \mathbf{S}(x, :) \in \mathbb{R}^N$, contains its dependencies on all key tokens. Since $N = N_h N_w N_b$, this row can be reshaped without information loss into a query-conditioned third-order tensor $\mathcal{G}_x \in \mathbb{R}^{N_h \times N_w \times N_b}$ with $\mathcal{G}_x(j_h, j_w, j_b) = S_{x,\pi(j_h,j_w,j_b)}$. This defines a bijective tensorization operation. It preserves every attention coefficient while restoring the original spatial–spectral arrangement of the key tokens. Therefore, $\mathcal{G}_x$ can be interpreted as a query-conditioned Tucker core whose entries are dynamically generated from query–key similarities rather than treated as free static parameters.

For each grid location (j_h, j_w, j_b), we define a separable tensor-valued feature as $\mathcal{V}_{j_h,j_w,j_b} = \mathbf{V}_1(:, j_h) \circ \mathbf{V}_2(:, j_w) \circ \mathbf{V}_3(:, j_b)$. The tensor-valued attention output associated with query token x is then given by

$$\begin{aligned}\mathcal{X}_x &= \sum_{j_h=1}^{N_h} \sum_{j_w=1}^{N_w} \sum_{j_b=1}^{N_b} \mathcal{G}_x(j_h, j_w, j_b) \mathcal{V}_{j_h,j_w,j_b} \\ &= \mathcal{G}_x \times_1 \mathbf{V}_1 \times_2 \mathbf{V}_2 \times_3 \mathbf{V}_3. \end{aligned} \tag{13}$$

Its element-wise expression is

$$\begin{aligned}\mathcal{X}_x(i_1, i_2, i_3) = &\sum_{j_h=1}^{N_h} \sum_{j_w=1}^{N_w} \sum_{j_b=1}^{N_b} S_{x,\pi(j_h,j_w,j_b)} \\ &\times V_1(i_1, j_h) V_2(i_2, j_w) V_3(i_3, j_b). \end{aligned} \tag{14}$$

Comparing (9) and (13), both operations perform weighted aggregation using dynamically computed attention coefficients. The distinction is that the value vectors in conventional self-attention are replaced by separable tensor-valued bases constructed from the three mode factor matrices. Under this parameterization, the Tucker multilinear product can be regarded as a tensor-valued extension of self-attention.

Based on this concept, the TA module is formulated as

$$\begin{aligned}\mathcal{X}_{\text{out}} &= \text{TA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3) \\ &= \mathcal{G} \times_1 \mathbf{V}_1 \times_2 \mathbf{V}_2 \times_3 \mathbf{V}_3 \in \mathbb{R}^{H \times W \times B}\end{aligned}$$

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$$\text{s.t. } \mathcal{G} = \text{Tensorize}\left[\text{SoftMax}\left(\frac{\mathbf{QK}^T}{\sqrt{d}}\right)\right]. \tag{15}$$

The tensorized attention core $\mathcal{G} \in \mathbb{R}^{N_h^2 \times N_w^2 \times N_b^2}$ is a blockwise rearrangement of the query-conditioned tensors $\{\mathcal{G}_x\}_{x=1}^{N}$ and defined as

$$\begin{aligned}&\mathcal{G}\left((i_h-1)N_h + j_h, (i_w-1)N_w + j_w, (i_b-1)N_b + j_b\right)\\ &= S_{\pi(i_h,i_w,i_b),\pi(j_h,j_w,j_b)}.\end{aligned} \tag{16}$$

Here, (i_h, i_w, i_b) specifies the spatial–spectral location of the query token, whereas (j_h, j_w, j_b) specifies the location of the corresponding key token. Therefore, $\mathcal{G}$ preserves all pairwise attention coefficients in $\mathbf{S} \in \mathbb{R}^{N \times N}$ while reorganizing them into a third-order tensor structure.

B. Sparse TA for HU

We now leverage the TA for the HU task. As discussed in (7), when the third-mode factor matrix is explicitly constrained to be the endmember matrix, the Tucker decomposition can be reformulated as a tensorized LMM. In (15), the size of $\mathbf{V}_3$ is $B \times N_b^2$ and the objective is to reconstruct the given tensor. However, in HU, the primary goal is to mine the underlying low-rank structure spanned by the endmembers, with a specific focus on the accuracy of the estimated abundances and endmembers.

Therefore, instead of utilizing the TA for HSI reconstruction, we repurpose it to explicitly guide abundance estimation. To achieve this, we decouple the third-mode projection into two matrices: a latent projection matrix $\mathbf{V}_3$ (mapping core features to abundances) and the endmember matrix $\mathbf{M}$ (mapping abundances to spectral bands). Consequently, the optimization problem in (15) is recast as

$$\begin{aligned}\min_{\mathbf{W}^Q, \mathbf{W}^K, \{\mathbf{V}_l\}_{l=1}^3, \mathbf{M}} \quad & \|\mathcal{Y} - \sigma\left(\mathcal{G} \times_1 \mathbf{V}_1 \times_2 \mathbf{V}_2 \times_3 \mathbf{V}_3\right) \times_3 \mathbf{M}\|_F^2\\ \text{s.t.} \quad & \mathcal{G} = \text{Tensorize}\left[\text{SoftMax}\left(\frac{\mathbf{QK}^T}{\sqrt{d}}\right)\right]\\ & \mathbf{Q} = \mathbf{PW}^Q, \quad \mathbf{K} = \mathbf{PW}^K\end{aligned} \tag{17}$$

where $\mathbf{V}_3 \in \mathbb{R}^{P \times N_b^2}$, and P denotes the number of endmembers. Here, $\sigma(\cdot)$ is the nonlinear activation function used to enforce physical constraints (e.g., nonnegativity and sum-to-one). In this formulation, the decomposition is treated as a two-step process: the first step maps the core tensor $\mathcal{G}$ to the abundance maps via $\{\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3\}$, and the second step reconstructs the hyperspectral tensor using the endmember matrix $\mathbf{M}$.

The advantages of applying the TA to unmixing are manifold. First, this module effectively captures local–global relationships within the HSI, attributed to the synergistic combination of local patch extraction and global dependency modeling. Second, when $N_h^2 < H$ or $N_w^2 < W$, the corresponding factor matrices directly induce a low-rank representation of the abundance maps. In contrast, when $N_h^2 > H$ or $N_w^2 > W$, the factor matrices primarily provide a compact low-rank representation of the spatial–spectral correlations among the extracted cubes. Such a representation effectively compresses the redundant nonlocal dependencies and facilitates the modeling of a compact nonlocal self-similarity prior. Consequently, the underlying abundance structure can be robustly recovered even in the presence of severe perturbations or noise.

To better tailor the mechanism for HU, we employ two task-specific modifications regarding similarity calculation and sparsity. In the standard attention mechanism, the dot product is used to measure similarity. However, in HSIs, the magnitude of spectral vectors may be distorted by illumination effects and topographic shadows, whereas the material identity is primarily encoded in the spectral shape. To mitigate the impact of intensity variations (spectral variability), we employ a cosine-similarity-based normalization

$$S_{xy} = \frac{\mathbf{q}_x^T \mathbf{k}_y}{\|\mathbf{q}_x\|_2 \ \|\mathbf{k}_y\|_2} \tag{18}$$

where $\mathbf{q}_x$ and $\mathbf{k}_y$ are the row vectors of $\mathbf{Q}$ and $\mathbf{K}$, respectively. This operation ensures that the attention weights reflect intrinsic material similarity rather than luminance differences.

Another fundamental premise in HU is that a specific local region is typically connected with only a small subset of all the regions in the scene. Traditional SoftMax activation produces a dense attention map, which contradicts this physical prior. To enforce sparsity explicitly, we replace SoftMax with SparseMax

$$\mathbf{S}_{\text{sparse}} = \text{SparseMax}\left(\mathbf{S}(x, :)\right). \tag{19}$$

The SparseMax function projects the input vector onto the probability simplex and is capable of returning exact zeros for low-probability elements. This results in a sparse attention map $\mathbf{S}_{\text{sparse}}$, which effectively filters out noise and irrelevant spectral signatures. Since the core tensor $\mathcal{G}$ is tensorized directly from $\mathbf{S}_{\text{sparse}}$, it inherently inherits this beneficial sparsity property.

C. Implementation Details

Given the core tensor $\mathcal{G}$ obtained from the tensorization of the attention map, we implement the decomposition described in (17) using the following concrete steps. First, we utilize a tensor contraction layer (TCL) [46] to generate the latent abundance features. This process is formulated as

$$\mathcal{Z} = \text{TCL}(\mathcal{G}) = \mathcal{G} \times_1 \mathbf{V}_1 \times_2 \mathbf{V}_2 \times_3 \mathbf{V}_3. \tag{20}$$

The resulting tensor $\mathcal{Z} \in \mathbb{R}^{H \times W \times P}$ represents the unconstrained latent abundance maps.

From a geometric perspective, under the LMM, each observed pixel lies within the convex hull spanned by the endmembers, and its abundance vector corresponds to the barycentric coordinates of that pixel in the endmember simplex. In our framework, SoftMax is used as a differentiable parameterization that maps the unconstrained latent variables onto this simplex. To this end, we implement $\sigma(\cdot)$ with a SoftMax function

$$\mathcal{A}_{ijk} = \frac{\exp\left(\mathcal{Z}_{ijk}\right)}{\sum_{p=1}^{P} \exp\left(\mathcal{Z}_{ijp}\right)}. \tag{21}$$

Here, $\mathcal{A} \in \mathbb{R}^{H \times W \times P}$ denotes the final estimated abundance map.

The second stage reconstructs the observed spectral signatures. We employ a linear transformation layer (without bias)

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to map the abundance vectors back to the observed spectral dimension B. Mathematically, this layer corresponds to the mode-3 product in the Tucker decomposition. The weight matrix of this linear layer, denoted as $\mathbf{W} \in \mathbb{R}^{P\times B}$, is interpreted as the transpose of the endmember matrix $\mathbf{M} \in \mathbb{R}^{B\times P}$. Finally, the reconstruction of the hyperspectral tensor $\widehat{\mathcal{Y}}$ is achieved via the linear combination of the endmembers weighted by their corresponding abundances

$$\widehat{\mathcal{Y}}(i,j,:) = \mathbf{M}\,\mathcal{A}(i,j,:)^T. \tag{22}$$

To estimate the optimal network parameters $\Theta = \{\mathbf{W}_Q, \mathbf{W}_K, \{\mathbf{V}_l\}_{l=1}^3, \mathbf{M}\}$, we employ gradient descent to minimize the reconstruction error. The objective function is defined as the Frobenius norm of the difference between the observed and reconstructed tensors

$$\mathcal{L}\left(\mathcal{Y}, \widehat{\mathcal{Y}}\right) = \left\|\mathcal{Y} - \widehat{\mathcal{Y}}\right\|_F^2. \tag{23}$$

To enforce the physical constraint that spectral reflectance must be nonnegative, we apply a projected gradient descent (PGD) step during the training process. Specifically, after each gradient update, the endmember matrix $\mathbf{M}$ is clamped to ensure all values remain positive

$$\mathbf{M} \leftarrow \max(\mathbf{M}, \epsilon) \tag{24}$$

where ϵ (set to 10^{-4}) is a small positive constant.

IV. Experiments and Analysis

In this section, we provide a detailed evaluation of the proposed method, including experimental results and comparative analysis.

A. Experimental Setting

1) Implementation: The proposed STU method was optimized using AdamW. A constant learning rate was set to 2.3×10^{-3} for the synthetic experiments and 3×10^{-3} for the real-world experiments. The model was trained for 5000 epochs in the main comparative experiments. For the convergence analysis only, the training process was extended to 12 000 epochs to better illustrate the reconstruction loss and unmixing metrics. All experiments were conducted on a workstation equipped with a single NVIDIA GeForce RTX 4090 GPU.

2) Baselines: To evaluate the effectiveness of the proposed STU method, we conducted comprehensive experiments on both synthetic and real-world hyperspectral datasets. We compared STU with seven other methods. These include the matrix-based TV-RSNMF [26], the tensor-based MV-NTF [30], and a set of deep learning approaches: uDAS [47], SRViT [41], DAEN [37], A2SAN [48], and SNMF-Net [42]. For the setup, we simply followed the parameter settings recommended in their original implementations.

3) Metrics: To quantify the unmixing performance, we measure the quality of the extracted endmembers and abundances against the reference ones with two metrics. The spectral angle distance (SAD) evaluates the shape similarity

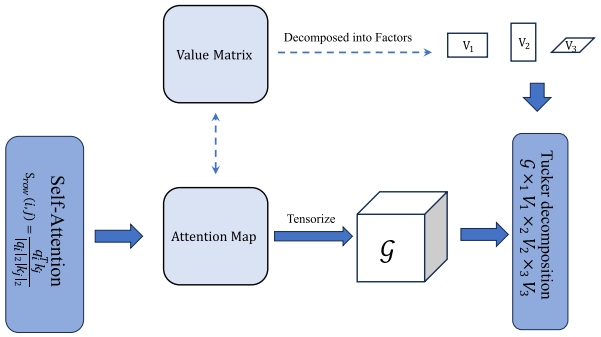


Fig. 2. Illustration of the Tucker decomposition and self-attention.

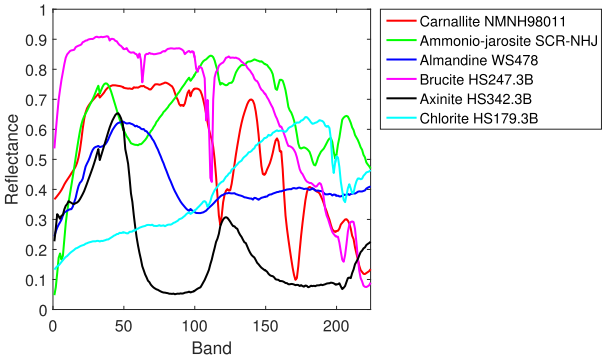


Fig. 3. Spectral signatures of six endmembers used in the synthetic data.

between the rth estimated endmember $\widehat{\mathbf{M}}_r$ and the reference endmember $\mathbf{M}_r$. It is defined as

$$\mathrm{SAD}_r = \arccos\left(\frac{\mathbf{M}_r^T\widehat{\mathbf{M}}_r}{\|\mathbf{M}_r\|_2\,\|\widehat{\mathbf{M}}_r\|_2}\right). \tag{25}$$

The root-mean-square error (RMSE) is employed to assess the reconstruction error of the abundance maps. Let $\mathbf{A}_r \in \mathbb{R}^{H\times W}$ denote the rth ground-truth abundance tensor and $\widehat{\mathbf{A}}_r$ denote the estimated one. The RMSE is calculated as

$$\mathrm{RMSE}_r = \sqrt{\frac{1}{HW}\left\|\mathbf{A}_r - \widehat{\mathbf{A}}_r\right\|_F^2}. \tag{26}$$

The average values of the SAD and RMSE are used to evaluate the unmixing performance.

B. Synthetic Data Generation

Following the protocol described in [42], we generated synthetic hyperspectral data. Six pure mineral signatures were selected: *Carnallite, Ammonio-jarosite, Almandine, Brucite, Axinite,* and *Chlorite*. These spectra cover 224 bands with wavelengths ranging from 0.38 to 2.5 μm. Their spectral signature is plotted in Fig. 3.

The abundance maps were generated through a spatially correlated process.

1) A synthetic image size of $z^2 \times z^2$ (e.g., 64×64) was partitioned into distinct nonoverlapping regions of size $z \times z$.
2) For each region, two endmembers were randomly selected from the library and initialized with mixing ratios θ and $1-\theta$, establishing a basic spatial pattern.

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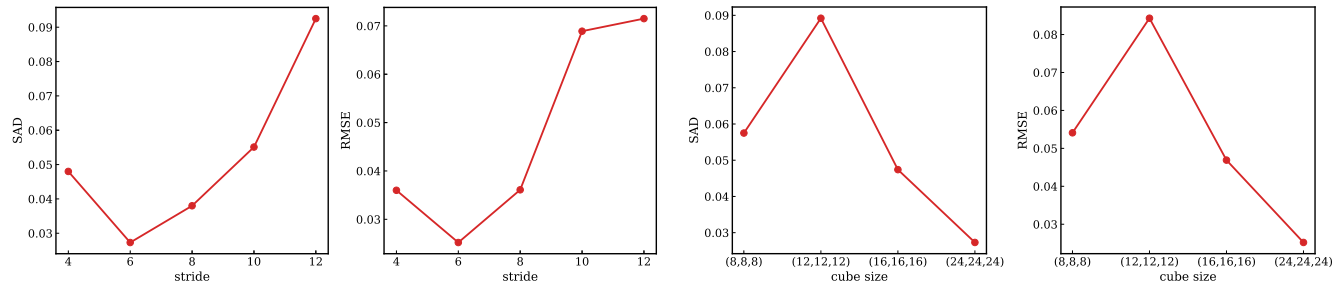


Fig. 4. Impact of (a) stride and (b) cube size with respect to the SAD and RMSE.

3) A low-pass Gaussian filter with a variance of 2 was applied to the initialized abundance maps.
4) The abundance fractions for each pixel were rescaled to satisfy the ASC.

To evaluate robustness, the clean synthetic data was corrupted with zero-mean additive Gaussian noise at different signal-to-noise ratios (SNRs), defined as

$$\mathrm{SNR} = 10\log_{10}\frac{\mathbb{E}\left[\mathbf{x}^T\mathbf{x}\right]}{\mathbb{E}\left[\mathbf{n}^T\mathbf{n}\right]} \tag{27}$$

where $\mathbf{x}$ and $\mathbf{n}$ represent the signal and noise vectors at a pixel, respectively.

C. Experiments on Synthetic Data

1) Impact of Cube Size and Stride: The cube size (K_h, K_w, K_b) defines the dimensions of the 3-D local volumes extracted by the sliding window, while the stride determines the step size of the window's movement. In this experiment, we evaluate the influence of both cube size and stride on unmixing performance using synthetic data with an SNR of 25 dB and $z = 8$.

First, we fixed the cube size at $(24, 24, 24)$ and varied the stride over $\{4, 6, 8, 10, 12\}$. The resulting SAD and RMSE values are shown in Fig. 4(a). The stride affects not only the number of extracted cubes and their overlap, but also the dimensions of the tensorized attention core, (N_h^2, N_w^2, N_b^2), and therefore directly controls the effective multilinear rank capacity of the proposed representation. A smaller stride generates more strongly overlapping cubes and richer redundant observations, which facilitates the exploitation of nonlocal self-similarity. However, an excessively small stride also produces a large number of highly correlated tokens, resulting in a high-dimensional tensorized correlation core and increased factor matrix sizes. This introduces optimization complexity. As the stride increases, the representation becomes more compact, and the unmixing performance initially improves, reaching its best value at $(6, 6, 6)$. When the stride becomes too large, however, the overlap and redundancy among neighboring cubes are substantially reduced, making the nonlocal spatial–spectral self-similarities less effectively exploited. Therefore, the stride of $(6, 6, 6)$ provides a favorable tradeoff among redundancy and representation complexity.

Next, with the stride fixed at $(6, 6, 6)$, we evaluated cube sizes of $(8, 8, 8)$, $(12, 12, 12)$, $(16, 16, 16)$, and $(24, 24, 24)$. The cube size also controls the dimensions (N_h^2, N_w^2, N_b^2) of the tensorized attention core through the number of extracted cubes. A smaller cube size produces a larger number of overlapping cubes and hence a higher-dimensional correlation core. Increasing the cube size reduces the number of cubes and the corresponding core dimensions, thereby providing a more compact multilinear representation of the spatial–spectral correlation context. Within the evaluated range, the cube size $(24, 24, 24)$ achieves the best unmixing performance and is therefore adopted in the subsequent experiments.

TABLE I
MEMORY CONSUMPTION (MB) UNDER DIFFERENT CUBE SIZE AND STRIDE SETTINGS

Cube Size	Stride				
	4	6	8	10	12
$8 \times 8 \times 8$	6794.70	705.03	212.94	–	–
$12 \times 12 \times 12$	5728.22	676.01	252.94	181.20	154.44
$16 \times 16 \times 16$	5481.78	1186.94	757.47	671.72	671.42
$18 \times 18 \times 18$	4972.32	1624.86	1331.57	1329.19	1328.38
$24 \times 24 \times 24$	9736.08	7331.52	7324.11	7321.72	7320.88

We also present a comprehensive analysis of memory consumption across varying joint configurations in Table I. When the cube size is fixed, a larger stride yields fewer tokens, dramatically lowering peak GPU memory consumption. However, as noted in the article, setting the stride too high degrades unmixing accuracy because it leaves too little patch overlap to properly exploit nonlocal self-similarities. When the stride is fixed, expanding the cube dimensions inherently increases the dimensionality of each token vector. This inflates memory demands during tokenization. Our chosen configuration of a $(24, 24, 24)$ cube size paired with a stride of 6 balances these demands, maintaining high accuracy within typical hardware capacity limits.

2) Convergence Behavior: We track the progression of the loss function alongside the unmixing performance metrics over training. As illustrated in Fig. 5, the reconstruction objective and the accuracy metrics cleanly converge and plateau well before the final epoch, verifying training stability.

3) Comparison Under Different Mixing Levels: Higher degrees of spectral mixing inherently increase the complexity of the unmixing task, as the separation between endmember signatures becomes less distinct. To evaluate the robustness of our method against varying mixing conditions, we varied the

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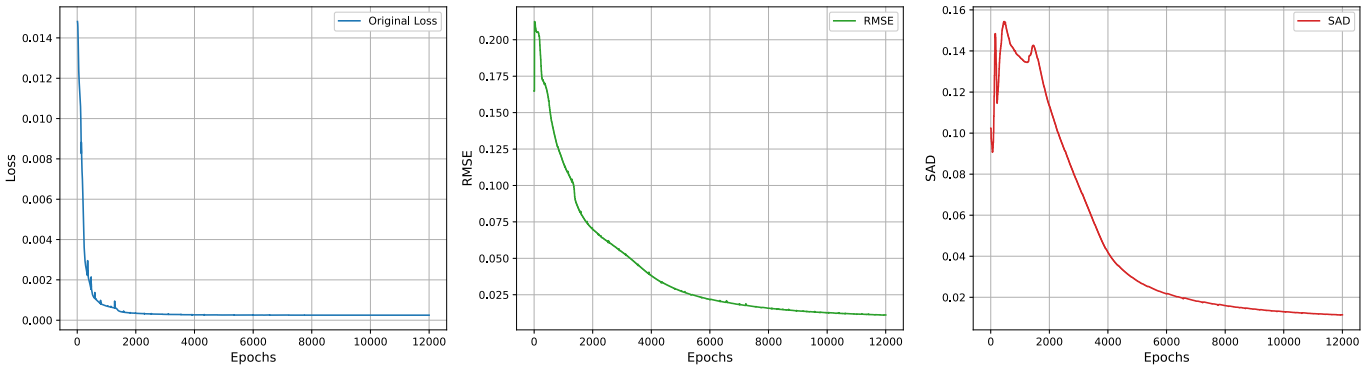


Fig. 5. Changes in the loss and unmixing performance during training.

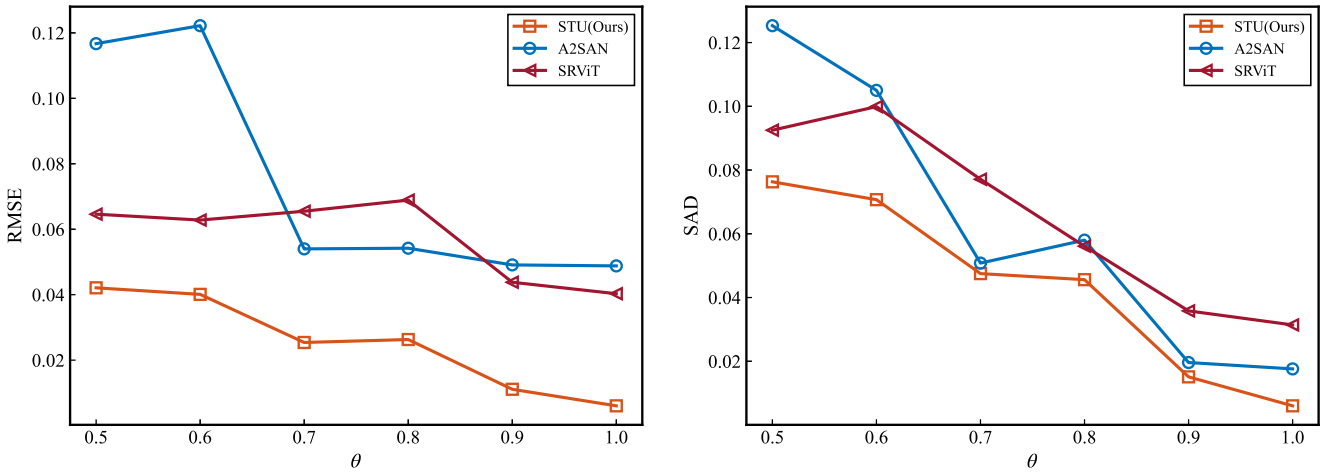


Fig. 6. Performance of STU, SRViT, and A2SAN under different mixing levels in terms of (a) RMSE and (b) SAD.

TABLE II
COMPARISON OF MODEL PARAMETERS AND RUNNING TIME

Algorithm	Parameters	Time
A2SAN	47,398	107s
SRViT	44,430,000	90s
STU (Ours)	1,793,944	105s

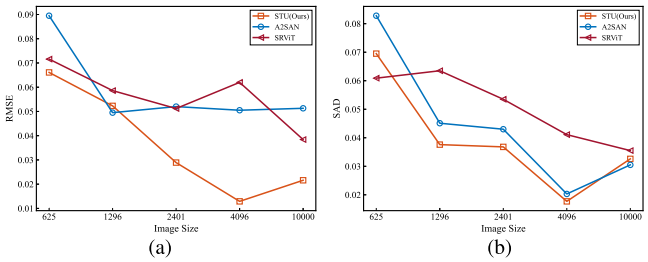


Fig. 7. Performance of STU, SRViT, and A2SAN under different image sizes in terms of (a) RMSE and (b) SAD.

mixing parameter θ from 0.5 to 1.0 with a step size of 0.1. For comparison, we included A2SAN and SRViT that employ spatial–spectral attention for unmixing.

As illustrated in Fig. 6, our proposed method consistently outperforms A2SAN and SRViT across all mixing levels. This superior performance is attributed to the sparse TA, which effectively captures local–global spatial–spectral correlations. Unlike standard attention mechanisms used in SRViT and A2SAN, the sparse TA enforces a robust low-rank structure derived from nonlocal self-similarities, which serves as a universal constraint suitable for all HSIs.

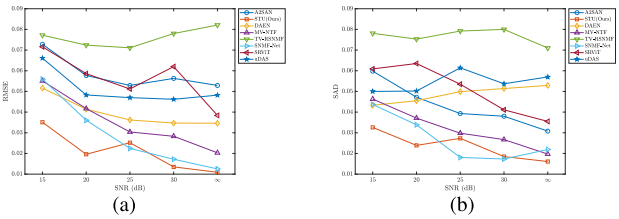
To comprehensively assess model complexity and computational cost, we have added a dedicated efficiency evaluation. As all the methods are unsupervised, the computational cost is measured by the training time. As detailed in Table II, our proposed STU achieves a massive reduction in model parameters compared to the standard Transformer-based baseline, SRViT (1.8M versus 44.4M parameters). Furthermore, our model maintains a highly competitive execution speed (105 s).

4) Comparison Under Different Image Sizes: Larger images typically introduce more complex spatial structures and provide a greater number of samples for the unmixing. To systematically evaluate the impact, we varied the image size across the set $\{25 \times 25, 36 \times 36, 49 \times 49, 64 \times 64, 100 \times 100\}$. As illustrated in Fig. 7, our method achieves performance comparable to A2SAN in terms of SAD. However, regarding the

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TABLE III
MEANS AND STANDARD DEVIATIONS OF THE SAD ON URBAN DATA

Algorithm	TV-RSNMF [26]	MV-NTF [30]	DAEN [37]	uDAS [47]	A2SAN [48]	SRViT [41]	SNMF-Net [42]	STU
Asphalt	**0.1028±1.17%**	0.1720±0.75%	1.0224±0.27%	0.2174±0.04%	0.1095±0%	0.1483 ±1.61%	0.2477±0.93%	0.1805±2.04%
Grass	0.1463±1.40%	0.1974±2.97%	0.5122±0.29%	0.6862±33.88%	0.1466±0%	0.1482±2.13%	0.2131±2.81%	**0.1315±2.09%**
Tree	0.2628±1.72%	0.1284±0.32%	**0.0801±0.01%**	0.0955±1.12%	0.0971±0%	0.085±0.98%	0.1848±2.05%	0.0867±1.16%
Roof	0.3332±8.00%	0.3608±3.66%	0.1069±0.03%	0.2037±2.37%	0.3338±0%	0.1487±7.02%	**0.0554±2.56%**	0.1067±1.83%
Mean	0.2113±1.04%	0.2147±0.96%	0.4304±0.02%	0.3007±8.79%	0.1717±0%	0.1325±1.15%	0.1753±0.55%	**0.1263±1.53%**



(a) (b)

Fig. 8. Performance of all methods under different noise conditions in terms of (a) RMSE and (b) SAD.

RMSE, our approach demonstrates a significant performance advantage.

5) Robustness to Noise: To evaluate the robustness of our method against varying noise conditions, we tested SNRs of 15, 20, 25, and 30 dB, and a noise-free baseline. The quantitative results are presented in Fig. 8. SNMF-Net achieves competitive performance, largely due to its effective integration of physical priors derived from the LMM, the alternating optimization, and sparse representation theory. However, our proposed STU method consistently outperforms SNMF-Net across most scenarios, with the performance gap becoming particularly pronounced under high-noise conditions. This superior robustness is primarily attributed to the sparse tensor prior module. By employing SparseMax instead of Softmax, the module generates significantly sparser attention maps and effectively filters out noise and irrelevant spectral signatures. This is helpful to more accurately capture the spatial–spectral correlation within HSIs. By further projecting the core tensor to produce abundance, our STU is more capable of recovering clean, structured regions even in the presence of heavy noise.

D. Experiments on Real-World Data

We then evaluate all the methods on the real-world Urban, Jasper Ridge, and Washington DC datasets. Because of the lack of a ground-truth abundance map, we qualitatively evaluate all the methods.

1) Urban Dataset: The Urban dataset, captured by the HYDICE sensor, is a widely used benchmark for real-world unmixing evaluation. The original image consists of 307 × 307 pixels with 210 spectral bands covering the range of 0.4–2.5 μm. After removing bands affected by water absorption and atmospheric noise, 162 informative bands remain. The scene contains four predominant endmembers: asphalt, grass, tree, and roof.

Table III presents the quantitative comparison of all competing methods in terms of the SAD metric. As observed, the proposed STU achieves the best overall performance, yielding the lowest average SAD across all materials. Figs. 9 and 10 illustrate the extracted endmember signatures and the corresponding abundance maps, respectively. Visual inspection confirms that our method yields the closest match to the reference ones among all compared approaches. This superior performance stems from the sparse TA's inherent ability to capture local–global context and effectively model spatial–spectral low-rankness.

2) Jasper Ridge Dataset: The Jasper Ridge dataset is one of the widely used benchmark datasets for HU and land-cover classification. It was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and initially consists of 512 × 614 pixels with 224 spectral bands covering the wavelength range from 0.38 to 2.5 μm. For experimental evaluation, a subimage of size 100 × 100 pixels is cropped from the original scene. Following standard practice, spectral bands heavily affected by water vapor absorption and atmospheric noise (bands 1–3, 108–112, 154–166, and 220–224) are removed, resulting in 198 clean spectral channels for unmixing. The Jasper Ridge scene contains four dominant material classes: *tree, water, soil,* and *road*.

As shown in Table IV, the proposed STU method achieves competitive performance and slightly underperforms SNMF-Net, which benefits from the stronger initialization provided by L1/2-NMF [23]. It is worth noting that our method only employs VCA for endmember initialization, without relying on more sophisticated prior initialization strategies. Under this setting, STU still ranks second among all compared methods, demonstrating its strong unmixing capability and robustness on real hyperspectral datasets. We have also added visual comparisons of the recovered endmember signatures and abundance maps in Figs. 11 and 12, respectively. The endmembers estimated by STU closely follow the corresponding reference spectral signatures, while the recovered abundance maps exhibit clear spatial structures and strong consistency with the reference distributions.

3) Washington DC Mall Dataset: The Washington DC Mall dataset was also acquired by the HYDICE sensor. The original image spans spatial dimensions of 1208 × 307 pixels and contains 210 spectral bands covering the wavelength range from 0.4 to 2.4 μm. For our experiments, a subimage of size 150 × 150 pixels was cropped from the original scene. Following standard preprocessing procedures, spectral bands affected by water absorption and severe atmospheric noise were removed, resulting in 191 effective bands for unmixing. In this study, five primary endmembers are considered: *tree, grass, street, roof,* and *water*.

Table V presents the quantitative comparison of all methods in terms of the SAD. Notably, the proposed STU method achieves the best performance on the tree and street

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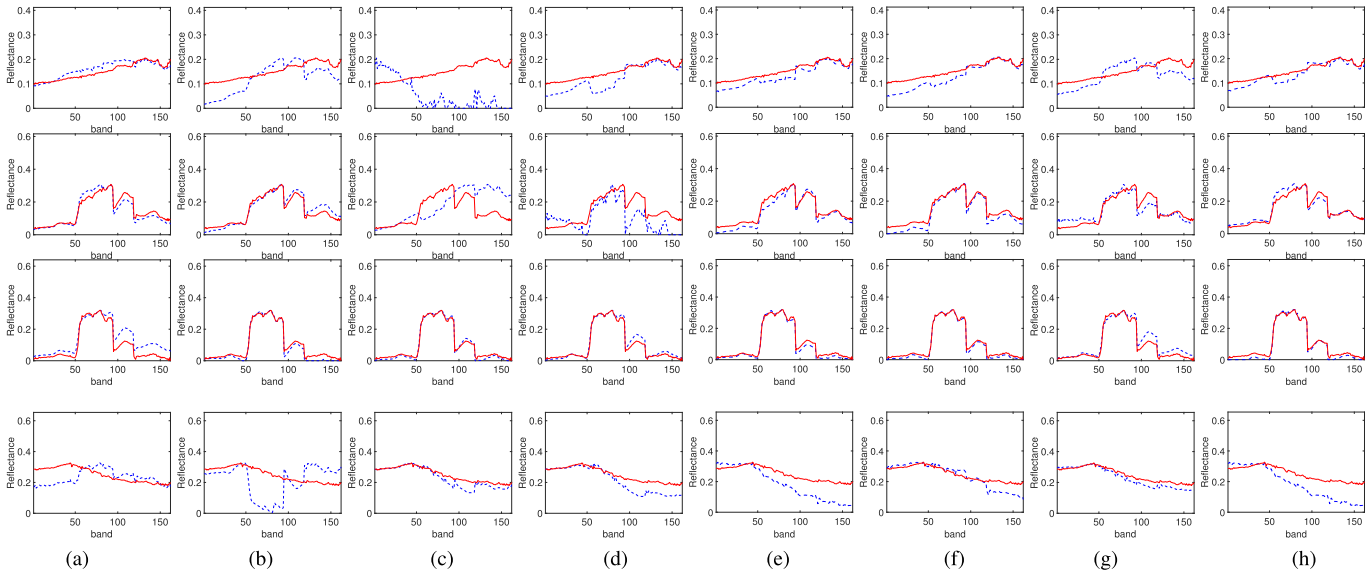


Fig. 9. Estimated endmembers by eight unmixing algorithms on the HYDICE Urban dataset. From top to bottom, the rows are the spectral signatures of asphalt, grass, tree, and roof, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2SAN. (f) SRViT. (g) SNMF-Net. (h) STU.

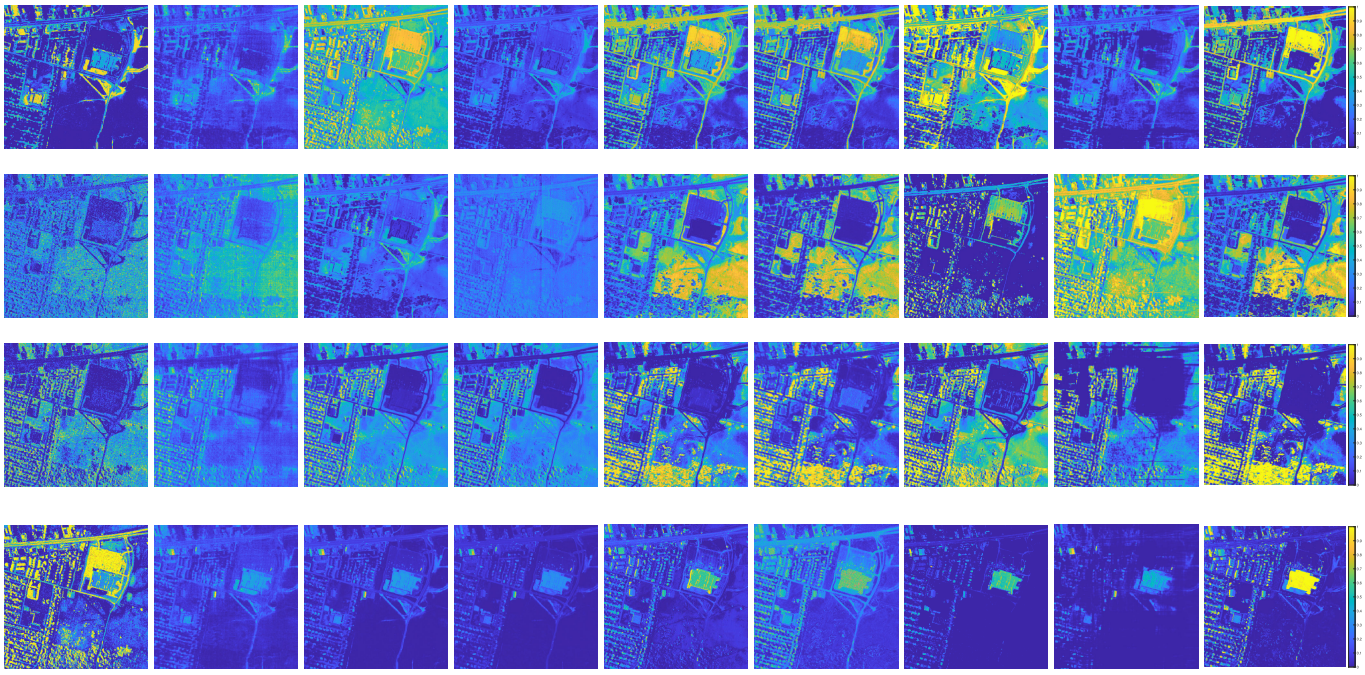


Fig. 10. Estimated abundance maps by eight unmixing algorithms on the HYDICE Urban dataset. From top to bottom, the rows are the abundance maps of asphalt, grass, tree, and roof, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2SAN. (f) SRViT. (g) SNMF-Net. (h) STU. (i) Reference.

TABLE IV
MEANS AND STANDARD DEVIATIONS OF THE SAD ON JASPER RIDGE DATASET

Algorithm	TV-RSNMF [26]	MV-NTF [30]	DAEN [37]	uDAS [47]	A2SAN [48]	SRViT [41]	SNMF-Net [42]	STU
Tree	0.0776±4.91%	0.2209±2.17%	0.1774±0.82%	0.1511±1.37%	0.1274±0%	0.1049±4.35%	**0.0478±1.62%**	0.0597±1.29%
Water	0.1337±1.31%	0.2352±1.65%	0.3237±1.42%	0.1742±11.39%	0.1627±0%	**0.0708±0.06%**	0.0855±1.77%	0.1101±0.56%
Soil	0.1047±10.11%	0.1752±4.67%	0.1123±2.03%	0.1639±6.09%	0.0957±0%	0.1450±1.40%	0.0443±3.09%	**0.0313±1.71%**
Road	0.6515±27.46%	0.1741±4.35%	**0.0588±1.59%**	0.0597±0.27%	0.1519±0%	0.1909±3.99%	0.0875±0.05%	0.0874±0.29%
Mean	0.2419±3.06%	0.2015±1.77%	0.1680±0.43%	0.1373±3.27%	0.1344±0%	0.1279±1.58%	**0.0663±1.04%**	0.0721±0.26%

endmembers, as well as the lowest average SAD across all materials. Figs. 13 and 14 illustrate the extracted endmember signatures and corresponding abundance maps. As expected, our STU method also presents good performance. This

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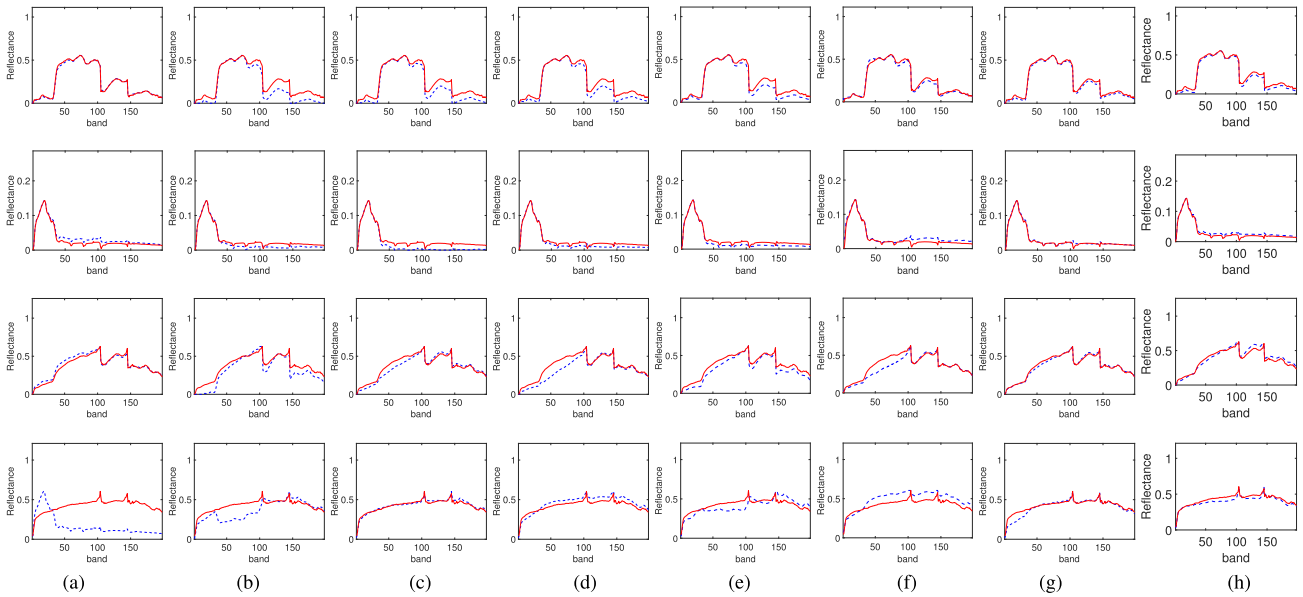


Fig. 11. Estimated endmembers by eight unmixing methods on the Jasper Ridge dataset. From top to bottom, the rows are the spectral signatures of tree, water, soil, and road, respectively. Solid lines denote the reference endmembers, and dashed lines denote the estimated endmembers. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2SAN. (f) SRViT. (g) SNMF-Net. (h) STU.

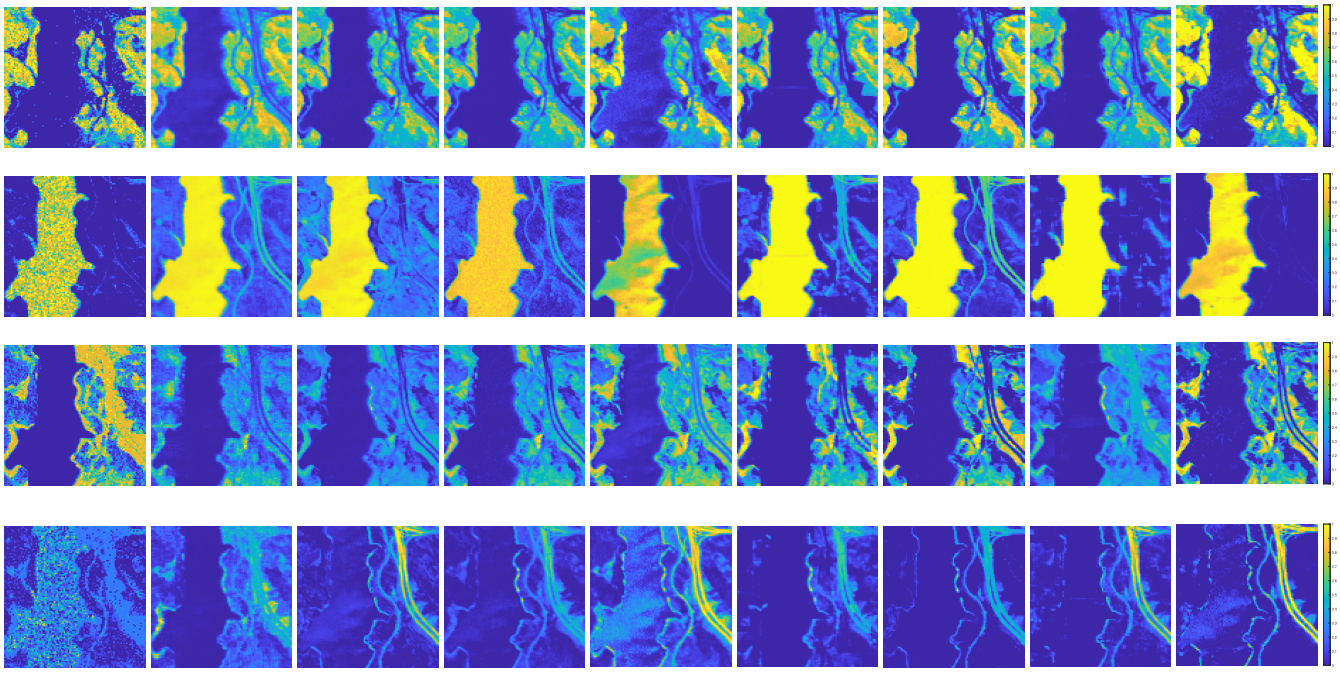


Fig. 12. Estimated abundance maps by eight unmixing methods on the Jasper Ridge dataset. From top to bottom, the rows are the abundance maps of tree, water, soil, and road, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2SAN. (f) SRViT. (g) SNMF-Net. (h) STU. (i) Reference.

TABLE V

MEANS AND STANDARD DEVIATIONS OF THE SAD ON WASHINGTON DC MALL DATA

Algorithm	TV-RSNMF [26]	MV-NTF [30]	DAEN [37]	uDAS [47]	A2SAN [48]	SRViT [41]	SNMF-Net [42]	STU
Tree	0.2071±1%	0.1211±0.53%	0.1378±2.33%	0.0939±2.15%	0.0887±0%	0.1276±0.34%	0.1309±1.06%	**0.0564±1.64%**
Grass	0.2961±4.34%	0.2357±0.40%	0.3773±17.84%	0.2888±6.03%	**0.1346±0%**	0.1604±0.78%	0.2180±2.09%	0.1884±2.07%
Street	0.3338±1.57%	0.4183±0.75%	0.3870±17.49%	0.3270±10.34%	0.3689±0%	0.3904±0.93%	0.3836±2.10%	**0.2835±3.20%**
Roof	0.2163±2.84%	0.0818±3.46%	0.1855±10.50%	0.0801±3.06%	0.1328±0%	0.036±0.25%	**0.0356±0.51%**	0.1157±1.54%
Water	**0.0466±0.53%**	0.1317±0.70%	0.1760±5.88%	0.1264±9.28%	0.0835±0%	0.0966±4.42%	0.0602±1.70%	0.0744±1.06%
Mean	0.2200±0.51%	0.1977±0.80%	0.2527±3.42%	0.1833±1.99%	0.1617±0%	0.1622±0.77%	0.1657±0.63%	**0.1436 ±1.15%**

superior performance can be attributed to two key components of the STU framework. On the one hand, the TA mechanism captures long-range dependencies across the entire scene, allowing the model to distinguish spectrally similar

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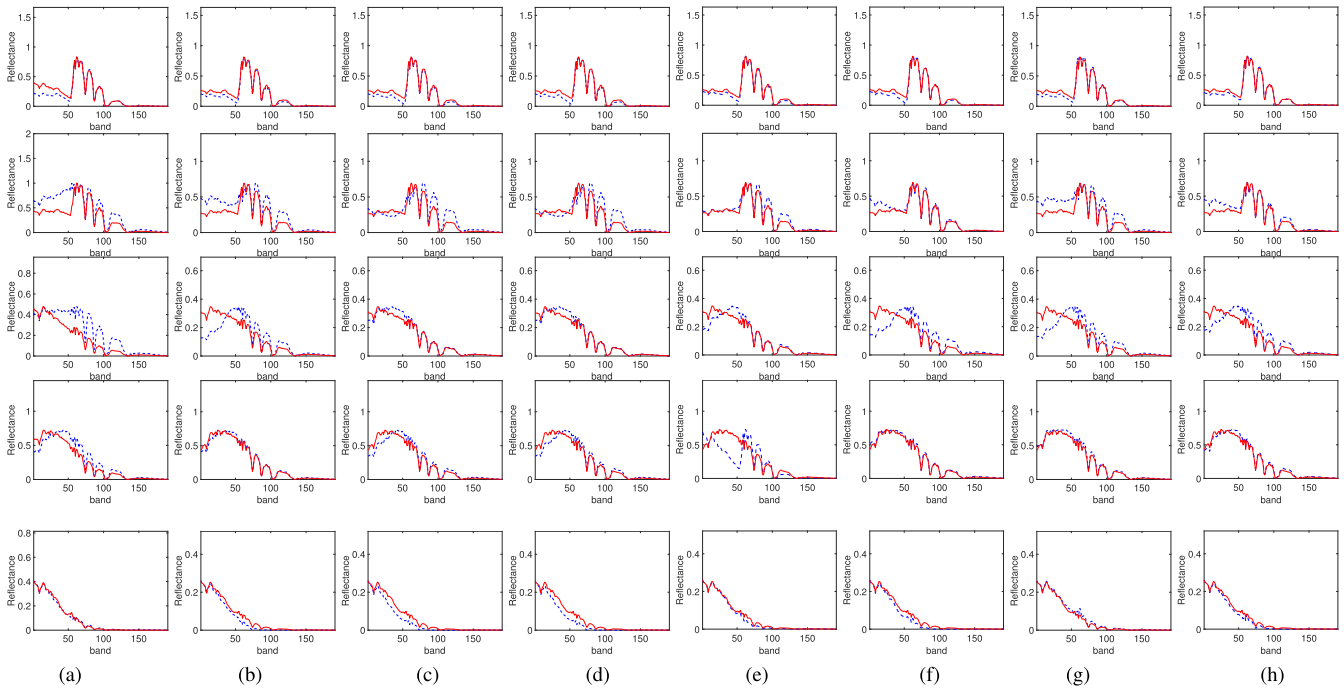


Fig. 13. Estimated endmembers by eight unmixing algorithms on the Washington DC Mall data. From top to bottom, the rows are the spectral signatures of tree, grass, street, roof, and water, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2SAN. (f) SRViT. (g) SNMF-Net. (h) STU.

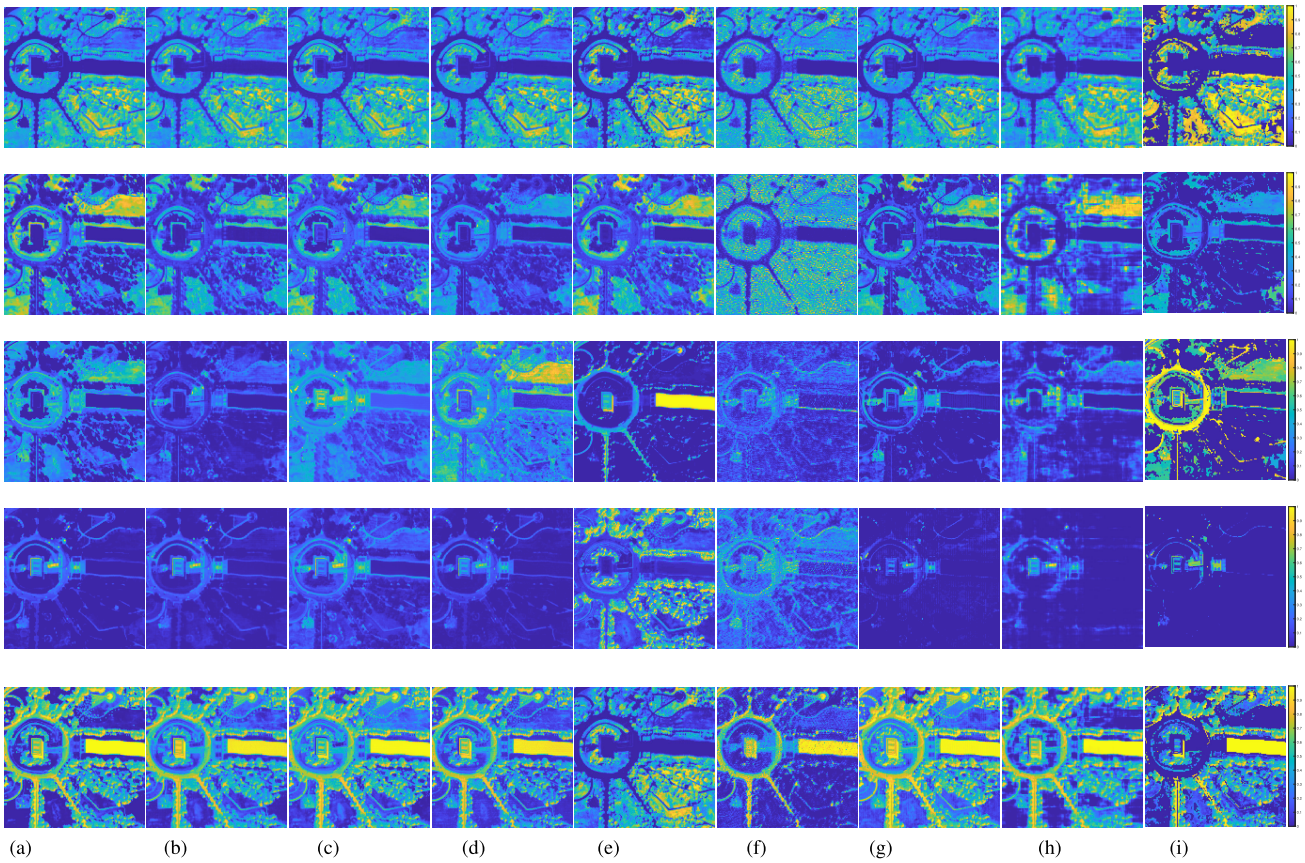
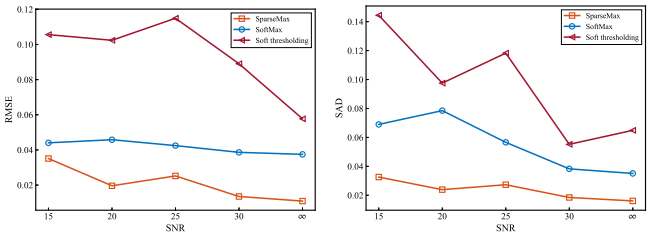


Fig. 14. Estimated abundance maps by eight unmixing algorithms on the Washington DC Mall data. From top to bottom, the rows are the abundance maps of tree, grass, street, roof, and water, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2SAN. (f) SRViT. (g) SNMF-Net. (h) STU. (i) Reference.

materials (like tree versus grass) by leveraging global contextual information. On the other hand, the integration of the SparseMax activation within the tensor core ensures that the learned attention maps remain sparse, resulting

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(a) (b)

Fig. 15. Ablation study on the SparseMax function versus soft thresholding and Softmax function. (a) RMSE. (b) SAD.

in more accurate endmember signatures and abundance maps.

E. Ablation Study

Here, we conducted an empirical comparison by replacing the SparseMax function with a normalized soft-thresholding operator. Mathematically, the alternative sparse attention map $\mathbf{S}_{\text{soft}}$ is calculated by

$$\mathbf{S}_{\text{soft}} = \text{Normalize}\left(\max\left(0, \frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}} - \lambda\right)\right) \tag{28}$$

where λ is a learnable thresholding parameter, $\max(0, \cdot)$ represents the ReLU-based soft-thresholding operation, and Normalize$(\cdot)$ performs row-wise L_1 normalization to ensure that the total weights sum to one.

We compared the unmixing performance of this soft-thresholding baseline against our proposed SparseMax implementation on the synthetic dataset, keeping all other hyperparameters completely identical. As shown in Fig. 15, the use of SparseMax consistently yields superior unmixing performance. We attribute the superiority of SparseMax over normalized soft-thresholding to the following reason. Soft-thresholding followed by normalization is a heuristic, two-step procedure. The thresholding step creates zeros, but the subsequent normalization step alters the relative differences and scaling of the remaining active elements. In contrast, SparseMax is derived as a constrained optimization problem that solves the Euclidean projection onto the simplex in a single unified step. This preserves the exact gradient structure of the active coordinates during backpropagation.

We also conducted additional ablation studies by replacing the proposed cosine similarity and SparseMax with the standard dot-product similarity and Softmax, respectively. As can be observed in Fig. 15, removing cosine similarity and SparseMax consistently leads to degraded unmixing performance. The main reason is that cosine similarity encourages the proposed STU framework to focus more on the spectral curve shape rather than absolute intensity values when measuring spatial–spectral relationships, thereby improving robustness to spectral variability. In contrast, standard dot-product similarity is more sensitive to magnitude variations and noise interference.

Moreover, the sparsity-inducing property of SparseMax helps suppress inaccurate relationships caused by noise and irrelevant feature interactions, resulting in cleaner and more discriminative representations. Consequently, the combination of cosine similarity and SparseMax contributes significantly to the overall effectiveness and robustness of the proposed method.

V. Conclusion

In this article, we proposed a novel self-supervised HU framework driven by sparse TA, termed STU. Unlike conventional deep-learning-based methods that rely on predetermined network architectures and implicitly encoded priors, STU constructs an adaptive tensor representation from the intrinsic spatial–spectral structure of the observed HSI. Building upon a mathematical connection between Tucker decomposition and self-attention, the proposed method reorganizes nonlocal token similarities into a tensorized attention core, enabling the effective characterization of global spatial–spectral dependencies and multilinear low-rank structures. The resulting sparse tensor core is projected to obtain structured abundance estimates, which are subsequently coupled with a learnable endmember matrix under the LMM. Extensive experiments on synthetic and real-world hyperspectral datasets demonstrate that STU achieves competitive or superior performance to state-of-the-art methods. In future work, we will further investigate more robust initialization strategies and endmember-bundle-based modeling to alleviate the instability caused by the pure-pixel assumption of VCA initialization, especially in scenes with highly mixed pixels or strong endmember variability. In addition, we will explore token merging and pruning strategies to reduce the memory consumption caused by pairwise similarity computation. Moreover, we will extend the proposed tensor-attention paradigm to nonlinear and spectrally variable mixing scenarios, as well as to other hyperspectral inverse problems, including super-resolution and denoising.

References

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[3] F. Xiong, Z. Wu, J. Zhou, S. Jia, and Y. Qian, "SUIT: Spatial–spectral union-intersection interaction network for hyperspectral object tracking," *IEEE Trans. Image Process.*, vol. 34, pp. 7786–7800, 2025.

[4] N. Li, S. Jiang, Y. Zhou, W. He, and S. Jia, "Fuzzy boundary-aware network for hyperspectral individual tree fine recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5541016.

[5] S. Zhang et al., "SAGT: Structure-adaptive graph transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5534416.

[6] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[7] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[8] F. Xiong, "Overparameterized nonnegative tensor factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5509113.

[9] B. Rasti, A. Zouaoui, J. Mairal, and J. Chanussot, "Image processing and machine learning for hyperspectral unmixing: An overview and the HySUPP Python package," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517631.

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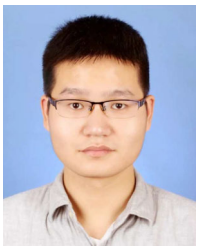
[10] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.
[11] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.
[12] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–277, Oct. 1999.
[13] X. Fu, K. Huang, B. Yang, W.-K. Ma, and N. D. Sidiropoulos, "Robust volume minimization-based matrix factorization for remote sensing and document clustering," *IEEE Trans. Signal Process.*, vol. 64, no. 23, pp. 6254–6268, Dec. 2016.
[14] A. Ambikapathi, T.-H. Chan, W.-K. Ma, and C.-Y. Chi, "Chance-constrained robust minimum-volume enclosing simplex algorithm for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4194–4209, Nov. 2011.
[15] J. Li, A. Agathos, D. Zaharie, J. M. Bioucas-Dias, A. Plaza, and X. Li, "Minimum volume simplex analysis: A fast algorithm for linear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5067–5082, Sep. 2015.
[16] S. Zhang, A. Agathos, and J. Li, "Robust minimum volume simplex analysis for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6431–6439, Nov. 2017.
[17] D. C. Heinz and Chein-I-Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.
[18] R. Heylen, D. Burazerovic, and P. Scheunders, "Fully constrained least squares spectral unmixing by simplex projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4112–4122, Nov. 2011.
[19] Z. Shi, T. Shi, M. Zhou, and X. Xu, "Collaborative sparse hyperspectral unmixing using l_0 norm," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5495–5508, Sep. 2018.
[20] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.
[21] X.-R. Feng, H.-C. Li, R. Wang, Q. Du, X. Jia, and A. Plaza, "Hyperspectral unmixing based on nonnegative matrix factorization: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4414–4436, 2022.
[22] Z. Yang, G. Zhou, S. Xie, S. Ding, J.-M. Yang, and J. Zhang, "Blind spectral unmixing based on sparse nonnegative matrix factorization," *IEEE Trans. Image Process.*, vol. 20, no. 4, pp. 1112–1125, Apr. 2011.
[23] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.
[24] S. Zhang et al., "Spectral–spatial hyperspectral unmixing using nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505713.
[25] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.
[26] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.
[27] H. Peng, T.-Z. Huang, X.-L. Zhao, W.-H. Wu, J. Lin, and T.-Y. Ji, "A unified data-aware fidelity and regularization learning paradigm for thick cloud removal of multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5536813.
[28] Q. Jiang, X.-L. Zhao, W.-H. Wu, W.-J. Zheng, and J. Lin, "An arbitrary mode-3 dimensional tensor-tensor product for tensor train decomposition from interaction perspective," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 8, pp. 7326–7338, Aug. 2025.
[29] J. Lin, L.-Y. Li, X.-L. Zhao, and H. Yang, "Robust multi-subspace tensor dictionary learning for multi-temporal image reconstruction," *Inverse Problems*, vol. 42, no. 7, Jul. 2026, Art. no. 075017.
[30] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1790, Dec. 2017.
[31] J. S. Bhatt and M. V. Joshi, "Deep learning in hyperspectral unmixing: A review," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 2189–2192.
[32] B. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Blind hyperspectral unmixing using autoencoders: A critical comparison," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1340–1372, 2022.
[33] C. Zhao et al., "CResDAE: A deep autoencoder with attention mechanism for hyperspectral unmixing," *Remote Sens.*, vol. 17, no. 21, Oct. 2025, Art. no. 3622.
[34] G. Zhang, S. Mei, Y. Wang, H. Han, Y. Feng, and Q. Du, "Spectral variability-aware cascaded autoencoder for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5505812.
[35] L. Qi, F. Gao, J. Dong, X. Gao, and Q. Du, "SSCU-Net: Spatial–spectral collaborative unmixing network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5407515.
[36] W. Gao, J. Yang, Y. Zhang, Y. Akoudad, and J. Chen, "SSAF-Net: A spatial–spectral adaptive fusion network for hyperspectral unmixing with endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506015.
[37] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.
[38] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.
[39] Y. Duan, X. Xu, T. Li, B. Pan, and Z. Shi, "UnDAT: Double-aware transformer for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522012.
[40] Z. Yang, M. Xu, S. Liu, H. Sheng, and J. Wan, "UST-net: A U-shaped transformer network using shifted windows for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528815.
[41] Y. Su, L. Gao, A. Plaza, X. Sun, M. Jiang, and G. Yang, "SRViT: Self-supervised relation-aware vision transformer for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17585–17598, Oct. 2025.
[42] F. Xiong, J. Zhou, S. Tao, J. Lu, and Y. Qian, "SNMF-Net: Learning a deep alternating neural network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510816.
[43] Y. Qian, F. Xiong, Q. Qian, and J. Zhou, "Spectral mixture model inspired network architectures for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7418–7434, Oct. 2020.
[44] P. Chen, S. Li, L. Cheng, X. Fu, Y.-C. Wu, and S. Theodoridis, "FieldFormer: Self-supervised reconstruction of physical fields via tensor attention prior," *IEEE Trans. Signal Process.*, vol. 73, pp. 2704–2718, 2025.
[45] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.
[46] J. Kossaifi, A. Khanna, Z. Lipton, T. Furlanello, and A. Anandkumar, "Tensor contraction layers for parsimonious deep nets," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1940–1946.
[47] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.
[48] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.



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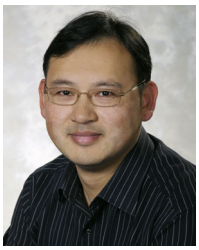


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